

DMBI Module 1

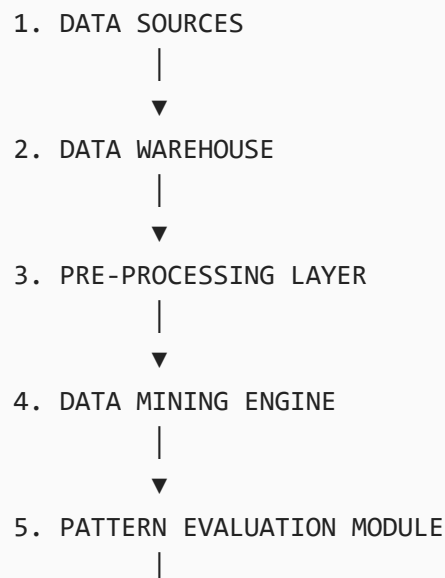
Define data mining. Explain the KDD process with a suitable diagram.

Data Mining is the process of discovering meaningful patterns and insights from large datasets using statistical, machine learning and computational techniques. It helps organizations analyse historical data and make data-driven decisions.

Knowledge Discovery in Databases (KDD) is the overall process of extracting meaningful knowledge from large datasets. Data Mining is one step within KDD, focusing on applying algorithms to discover patterns. KDD ensures raw data is transformed into valid, useful, and understandable knowledge.

Steps in KDD Process:

1. Data Selection: Choose relevant data from multiple sources (databases, warehouses, files).
2. Data Cleaning & Pre-processing: Remove noise, handle missing values, correct inconsistencies.
3. Data Transformation / Reduction: Convert data into suitable formats (normalization, aggregation, dimensionality reduction).
4. Data Mining: Apply algorithms (classification, clustering, association rules) to discover patterns.
5. Pattern Evaluation: Assess discovered patterns for validity, usefulness, and novelty.
6. Knowledge Representation: Present results using visualization, reports, rules, or charts for decision-making.



KDD Architecture Layers:

1. Data Sources: Operational databases, flat files, external sources.
 2. Data Warehouse: Central repository integrating data from multiple sources.
 3. Pre-processing Layer: Cleaning, integration, transformation of raw data.
 4. Data Mining Engine: Core algorithms (classification, clustering, regression, association).
 5. Pattern Evaluation Module: Validates interestingness of discovered patterns.
 6. Knowledge Presentation Layer: Visualization tools, reports, dashboards for end-users.
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What is a Decision Support System?

A computer-based system that supports decision-making using data and models

- Helps in solving semi-structured and unstructured problems
- Provides what-if analysis and forecasting
- Supports managers in making informed decisions

Example:

A DSS helps a manager decide the best pricing strategy by analysing demand and costs.